

UMEÅ UNIVERSITY

Designing Explainable Decision Support for Academic Study Progression in Higher Education

A Rule-Based Decision Support Approach to Study Progression and Understanding Degree Requirements

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Abstract

Higher education programs are governed by complex planning to comply with degree requirements that students must interpret when planning their studies. Although student information systems provide accurate records of completed courses, they offer limited support for understanding how these records relate to formal degree requirements, often leading to uncertainty about academic progression and eligibility.

This thesis investigates how a rule-based decision support system for course progression can be designed to ensure that students understand and trust its feedback regarding the fulfilment of degree requirements. The thesis adopts a decision support perspective in which formal academic regulations are explicitly represented using declarative rule-based reasoning, rather than relying on automated or data-driven decision-making approaches. Study planning is modelled as a combinatorial optimisation problem and implemented using Answer Set Programming to reason about prerequisites, credit accumulation, and degree constraints. The system generates multiple valid study trajectories and derives explanations directly from its reasoning process. These explanations are presented through an interactive interface that supports progressive disclosure, allowing users to access different levels of detail as needed.

The system is evaluated through a formative pilot study with a study counsellor and an exploratory user study with students. The results indicate that formally correct, transparently explained, and interactively accessible feedback supports student understanding and calibrated trust. Minimal explanations aid orientation, while detailed explanations are essential for resolving uncertainty and supporting informed decision-making. The thesis demonstrates that rule-based reasoning combined with explanation-driven interaction can provide effective decision support for academic study planning, complementing existing administrative systems and human advising without replacing them.

keywords: Decision support systems, academic progression, explainable systems, rule-based reasoning, Answer Set Programming, user trust, higher education

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1 Introduction

1.1 Problem statement

Higher education programs need complex planning to comply with degree requirements, including constraints on the total number of credits that must be completed, required academic level, which subject-areas the planned courses must fulfill, and under what additional restrictions these requirements apply. Students are expected to navigate these requirements as they plan their studies over several years, often choosing among a large course offering and make decisions that affect both their choice of specialization and the outcome of their education [6].

Although degree structures are formally defined and rule-based, it can be difficult to determine how individual course choices relate to degree requirements. This complexity is reflected in the organizational structure of academic advising system at Swedish universities, where each program is typically supported by specialized study counselors with expert knowledge of specific degree requirements and progression rules. According to the information provided by Umeå University, study counselors can assist students in all matters and questions related to studies, including degree regulations, eligibility, and long-term study planning [25]. The existence of such specialized counselor roles highlights that understanding and planning academic progression often requires interpretive support beyond what is provided solely by administrative information systems.

This difficulty is not primarily caused by a lack of information, but by the challenge of integrating information from multiple sources. Degree requirements are typically described in detailed program regulations published as documents or static web pages, while student progress data is presented separately in administrative systems. As a result, students must manually combine rules, course classifications, prerequisites, and accumulated credits to assess their own situation. This process can be cognitively demanding and particularly challenging for students who deviate from standard study plans, such as those who change focus, study part-time, or retake courses [9].

Supporting study planning and progression is therefore important, not only from an administrative perspective, but also from a student-centred one. Helping students understand their current academic status and future options has the potential to reduce uncertainty, improve decision-making, and increase students' sense of control over their education [6] and their outcomes. In practice, this need becomes especially pronounced in the later stages of education, where course choices become less structured and degree requirements must be satisfied through combinations of courses rather than predefined study paths.

Although it is technically possible to automate decisions such as course eligibility or graduation approval, academic planning is inherently a human-centred activity. Students' study choices are influenced by personal interests, workload preferences, career goals, and life circumstances that cannot be fully captured by formal rules alone. Research on human au-

tomation interaction has shown that high levels of automation can reduce user involvement and decision authority, particularly in complex, judgment-based tasks [18]. In the context of academic planning, fully automated decision-making therefore risks oversimplifying a complex decision space and reducing student agency.

Decision support systems offer an alternative approach by assisting users without replacing their judgment [7, 23]. Rather than making decisions on behalf of students, a decision support system can evaluate formal constraints, highlight potential issues, and present alternative study paths while leaving the final decision to the user. This distinction between support and automation is well established in decision support research and is particularly relevant in domains involving long-term and personally significant decisions.

In educational contexts, incorrect or misleading feedback can have serious consequences for a student's academic path, such as delayed graduation or enrolment in inappropriate courses. Correctness with respect to formal degree requirements is therefore a fundamental prerequisite for any system intended to support academic planning. However, correctness alone is not sufficient. If students cannot understand how a system reaches its conclusions, they may distrust its feedback or fail to use it effectively. Prior research has shown that a lack of explanation can undermine user trust and appropriate reliance on decision-supporting systems, even when those systems are technically accurate [20]. Conversely, systems that provide clear and comprehensible explanations can support users in forming an understanding of both the underlying rules and their own progression[1]. Explainability is thus closely tied to trust, perceived reliability, and effective use of decision support systems[10]. In the context of study planning, explainable feedback can help students reason about their options, identify potential issues early, and make informed decisions about their academic path.

Most universities provide student information systems that allow students to view their registered courses, completed credits, and academic results. In the Swedish higher education context, systems such as Ladok[1] serve as authoritative sources of academic records and administrative status. While such systems are essential for formal processes, they primarily focus on data presentation rather than decision support. Students can typically see what they have completed, but must often determine for themselves how this information relates to degree requirements, academic progression, and future eligibility. Information about degree requirements, specialization rules, and thesis eligibility is usually provided separately from administrative systems. In practice, these requirements are often described in program-specific regulations published as documents or static web pages on individual university websites. As a result, students must manually interpret textual requirement descriptions and relate them to their own study records in Ladok, a process that requires careful reading, cross-referencing, and familiarity with institutional terminology.

This thesis does not aim to replace study counsellors, existing administrative systems, or official requirement documentation. Instead, it explores how a complementary decision support system can assist students by making formal degree rules explicit, connecting them to individual study records, and presenting the resulting feedback in a way that is both correct and understandable. By doing so, such a system has the potential to support students in preparing for counseling meetings, reduce unnecessary uncertainty, and enable more informed discussions between students and advisors.

[1] `https://student.ladok.se/student/app/studentwebb/`

1.2 Research questions

Main Research Question:

> *How can a rule-based decision support system for course progression be designed so that students understand and trust its feedback regarding their fulfillment of degree requirements?*

To approach this main question, three sub-questions are introduced:

- Which rules and logical structures are required to accurately map students' course data against formal degree requirements?
- How do different forms of feedback presentation (minimal versus detailed explanations) influence students' understanding of their academic progression?
- How does the level of system transparency affect students' trust in the system and their perceived control over course planning?

Expected contributions and outcomes The thesis is expected to demonstrate that a rule-based decision support approach can provide formally correct and interpretable feedback on academic progression when degree requirements are explicitly modeled as logical rules. By representing academic planning as a constraint-based reasoning problem, the system is expected to support the generation of multiple valid study trajectories, reflecting the inherent flexibility of higher education programs and avoiding a single prescriptive notion of a "correct" study path. It is further expected that explanation design plays a critical role in how such feedback is perceived and used by students. Rather than assuming the existence of a single optimal presentation style, the study is expected to explore how different levels of explanation, ranging from minimal overview feedback to more detailed, support students' understanding of their academic situation in different ways. Finally, the thesis is expected to provide empirical evidence regarding how transparency and explainability influence students' trust in the system and their perceived control over course planning. Taken together, these expected outcomes position the system as a decision support tool that complements human academic advising by supporting understanding, exploration, and informed decision-making, rather than replacing personal judgment.

2 Background

This chapter presents the conceptual and technical foundations underlying the design choices made in this thesis. It introduces decision support systems, key principles from research on explainable systems and human–computer interaction, as well as relevant background on rule-based reasoning and planning as a combinatorial optimization problem. Together, these perspectives provide the necessary tools for understanding how the proposed system is designed and why its approach is appropriate for supporting students' understanding of academic progression.

2.1 Decision Support Systems

Decision support systems (DSS) are interactive computer-based systems that integrate data, analytical models, and user interaction to help users make informed decisions in complex problem domains. Rather than replacing human decision-making, a DSS augments it by providing structured information, analyses, or recommendations that help users reason about available options [24].

Bonczek, in his book "Foundations of Decision Support Systems," expresses that a defining characteristic of decision support systems is that they support, rather than replace, human judgment, particularly in complex decision situations [2]. This factor distinguishes DSS from automated decision-making systems, in which decisions are made entirely by the system without the user being involved. Prior research highlights that this difference is particularly important in human domains such as healthcare, law, and education, where decisions often have long-term consequences and depend on personal goals, contextual factors, or incomplete information [19]

Since the early 1970s, decision support systems have evolved and can be implemented using a variety of technical approaches, depending on the structure of the problem domain and the nature of the decisions being supported. Common approaches include procedural implementations based on hard-coded logic, rule-based systems, or constraint- and optimization-based formulations [2, 7].

Procedural approaches typically encode decision logic directly in program code. While such implementations can be efficient for narrowly defined problems, they often become difficult to maintain or extend as the number of rules and interactions grows, particularly when requirements change over time.

Rule-based approaches separate domain knowledge from control flow by representing requirements as explicit rules and constraints. This separation supports maintainability, traceability, and validation, particularly in domains governed by formal regulations [2].

Constraint-based and optimization-oriented approaches model decision-making problems

in terms of permissible and preferred solutions rather than fixed procedures. These approaches are well-suited to planning problems, where multiple valid solutions may exist and trade-offs must be evaluated.

2.2 Planning as a Combinatorial Optimization Problem

To understand how decision support can be provided for academic progression, it is necessary to characterize course planning as a problem domain. From a computational perspective, course planning can be viewed as a combinatorial optimization problem, in which a set of discrete decisions must satisfy a collection of constraints while optimizing one or more objectives.

From an artificial intelligence perspective, planning problems are commonly understood as the task of selecting actions that lead from an initial state toward desired goals while respecting a set of constraints. Such problems often admit multiple possible plans, which must be evaluated relative to stated objectives rather than treated as simply valid or invalid [5]. This view emphasizes planning as a form of structured reasoning over alternatives, rather than as the execution of a fixed procedure.

Combinatorial optimization provides a complementary computational formulation of this perspective. In such problems, decisions are drawn from a discrete search space, constraints define which combinations are permissible, and objective functions express preferences among feasible or partially feasible solutions [17]. Rather than yielding a single predetermined outcome, combinatorial optimization problems typically admit multiple valid solutions that differ in quality according to defined criteria.

In the context of academic planning, the decisions correspond to course selections distributed across academic terms. Constraints include prerequisites between courses, limits on the number of credits per term, and degree or area-specific requirements. At the same time, objectives may include maximizing progress toward degree completion, minimizing unmet requirements, or prioritizing certain categories of courses.

Importantly, not all planning situations admit a fully feasible solution at a given point in time. A student may, for example, be unable to satisfy all degree requirements due to missing prerequisites, limited course availability, or incomplete prior coursework. Treating course planning solely as a constraint satisfaction problem would, in such cases, result in failure rather than meaningful feedback. Optimization-based formulations address this limitation by allowing solutions to be evaluated relative to defined objectives, even when some constraints cannot yet be satisfied [21].

Framing course planning as a combinatorial optimization problem, therefore, enables the comparison and ranking of alternative study plans, rather than enforcing a binary distinction between valid and invalid outcomes. This perspective provides a principled basis for generating multiple possible trajectories, assigning scores to candidate plans, and identifying plans that represent the best available progress under the given constraints.

2.3 Rule-Based Reasoning and Answer Set Programming

2.3.1 Rule-based reasoning

Rule-based systems represent domain knowledge in the form of explicit rules that describe how conclusions can be derived from known facts. Reasoning proceeds by evaluating which rules apply in a given situation and by determining which conclusions follow from their interaction. This form of reasoning is particularly well-suited to domains ruled by formal regulations, where requirements are stated explicitly and can be interpreted as conditional constraints.

Academic degree requirements are typically defined by using structured rules, such as prerequisites, mandatory components, and minimum credit thresholds. These requirements are relatively stable and are intended to be applied consistently across students. Encoding such requirements as explicit rules enables systematic evaluation of whether a given set of completed or planned courses satisfies the conditions for graduation.

A central advantage of rule-based reasoning is the separation between domain knowledge and the mechanism that performs inference. The rules themselves specify what conditions must hold, while the reasoning process determines how conclusions are derived. This separation supports maintainability, traceability, and validation of the encoded knowledge, which is particularly important in educational contexts where incorrect interpretations may have significant consequences for students [5].

2.3.2 Declarative reasoning and Answer Set Programming

From a knowledge representation perspective, rule-based systems are commonly expressed using declarative formalisms. In declarative programming, the system designer specifies the properties that valid solutions must satisfy, rather than prescribing a sequence of computational steps to construct a solution. Declarative representations are especially appropriate for reasoning and planning tasks, where the primary concern lies in expressing constraints, alternatives, and goals rather than procedural control flow [5].

Answer Set Programming (ASP) is a declarative programming paradigm specifically designed for solving complex combinatorial problems involving many interacting rules and constraints. In ASP, a problem is described as a set of logical rules and constraints, and a solver computes one or more answer sets. Each answer set represents a solution that satisfies the given specification under the stable model semantics [12].

ASP has been established as a general-purpose framework for knowledge-intensive reasoning, with successful applications in planning, scheduling, configuration, and diagnosis. These problem classes share key characteristics with academic course planning, including discrete decision spaces, complex constraint interactions, and the existence of multiple valid solutions [3]

While these theoretical foundations establish the suitability of ASP as a paradigm, practical ASP systems also provide constructs that make such reasoning operational. Modern ASP solvers support expressive rule syntax, integrity constraints, and optimization statements, enabling concise representations of real-world planning problems. Tutorial-oriented treatments of ASP illustrate how answer sets correspond to alternative solutions and how constraints and preferences shape the solver's output in practice [8].

2.3.3 Constraints and optimization in ASP

A central concept in declarative reasoning, as discussed in Knowledge Representation, Reasoning, and the Design of Intelligent Agents [5], is the distinction between constraints that define acceptable states and preferences that guide the selection among acceptable alternatives. This distinction aligns closely with the formulation of planning problems as optimization problems rather than as purely feasibility-based tasks.

In ASP, this distinction is realized through the use of hard constraints and optimization statements. Hard constraints specify conditions that must be satisfied for an answer set to be considered valid, while optimization statements allow valid solutions to be ranked according to explicit criteria. This enables reasoning systems to compare alternative solutions rather than producing a single fixed outcome.

This capability is particularly important in educational planning contexts. At a given point in time, a student may be unable to satisfy all degree requirements due to missing prerequisites, limited course availability, or incomplete prior coursework. Treating all requirements as hard constraints would result in the absence of any solution and provide little meaningful feedback. By incorporating optimization, ASP allows solutions to be evaluated relative to defined objectives, such as minimizing unmet requirements or maximizing progress toward graduation.

Modern ASP systems support such optimization mechanisms directly within the declarative specification, allowing multiple answer sets to be generated and compared without embedding procedural control logic in the program. Tutorial treatments illustrate how optimization enables preference-based reasoning over alternative solutions in a transparent and systematic way [8].

Taken together, the use of explicit rule representations, declarative semantics, and integrated optimization makes Answer Set Programming a suitable framework for modeling academic course planning. It supports systematic reasoning over complex degree requirements, the generation of multiple alternative study trajectories, and the identification of best-possible plans under given constraints, directly reflecting the characteristics of course planning as a combinatorial optimization problem.

2.4 Design principles for explainable systems

2.4.1 From correct reasoning to understandable feedback

The rule-based and optimization-oriented reasoning framework described in the previous sections provides a principled way of determining whether academic requirements are fulfilled and of generating alternative study plans. However, correctness at the level of formal reasoning does not automatically translate into effective decision support. For a system to meaningfully support human decision-making, users must also be able to understand, evaluate, and trust the system's outputs.

This distinction is well established in decision support and human–computer interaction research. Systems that produce correct results but fail to communicate the reasoning behind those results risk being perceived as opaque or arbitrary, which may reduce user trust and limit effective use [11]. As a result, explainability must be treated as a core design concern

rather than as an optional add-on to system functionality.

In the context of academic planning, this issue is particularly salient. Students are expected to make long-term decisions based on system feedback, often in situations where requirements are complex and partially unmet. Providing explanations that help users reason about their progression is therefore essential for enabling informed and autonomous decision-making.

2.4.2 Explainability, transparency, and interpretability

Explainable system design involves several related but distinct concepts, most notably transparency, explainability, and interpretability. While these terms are often used interchangeably, they describe different aspects of how users relate to system outputs.

In recent years, these concepts have been increasingly discussed in the context of Trustworthy AI, where transparency and explainability are identified as key requirements for systems that support or influence human decision-making [4]. From this perspective, explainability is not only a technical concern, but a prerequisite for enabling human oversight, informed use, and appropriate reliance on system feedback.

Transparency refers to the extent to which a system's internal mechanisms are visible or accessible. In rule-based systems, transparency may be achieved by exposing rules, constraints, or intermediate reasoning steps. However, transparency alone does not guarantee that users can make sense of this information. Simply making internal logic visible does not ensure that it is meaningful or usable from a human perspective.

While **explainability** concerns the system's ability to communicate the reasons behind its outputs in a form that is meaningful to users. Explanations typically involve abstraction and selection, emphasizing relevant aspects of the reasoning process while excluding unnecessary internal detail [22]. Therefore, effective explanations bridge the gap between formal system logic and user understanding.

Interpretability describes the degree to which users can form an accurate mental model of how the system behaves. A system may be transparent in principle yet difficult to interpret if its explanations are poorly aligned with user goals or prior knowledge or contextual needs[15]. Interpretability is thus not an inherent effect of a system alone, but an emergent effect of the interaction between system, explanation, and user.

From a human-centered perspective, explainability should therefore be understood as a design outcome rather than a purely technical property. The objective is not to expose all internal reasoning, but to support user understanding at an appropriate level of abstraction, enabling informed judgment and calibrated trust [14].

2.4.3 Minimal and detailed explanations as a design dimension

One central design decision in explainable systems concerns the level of detail provided in explanations and feedback. Explanations can vary from minimal summaries of outcomes to detailed accounts of contributing factors and rule interactions.

Minimal explanations typically present high-level information, such as whether a requirement is fulfilled or how much progress remains. These kinds of explanations impose low cognitive load and allow users to quickly assess their situation, making them appropriate

for routine checks or users with existing domain familiarity.

Detailed explanations, by contrast, provide richer contextual information, such as which rules were applied, how individual courses contributed to a requirement, or why certain options were excluded. While more cognitively demanding, detailed explanations can support learning, error detection, and deeper understanding of complex reasoning processes [13].

Research in human–computer interaction suggests that the effectiveness of explanation depends on the user's goals, expertise, and the perceived importance of the decision[14]. Consequently, systems that support multiple explanation levels allow users to regulate the amount of information they receive, balancing efficiency with depth of understanding and supporting different forms of interaction over time [1].

2.4.4 Trust, perceived control, and decision support

Trust is a critical factor in the effective use of decision support systems. Users are unlikely to rely on system feedback if they do not trust its correctness, relevance, or limitations. Importantly, trust is shaped not only by objective accuracy but also by how system outputs are presented and explained.

Explanations play a key role in supporting calibrated trust, where users develop an appropriate level of reliance on the system rather than blindly accepting or dismissing its recommendations [11]. By understanding the reasoning behind system feedback, users are better able to judge when the system's suggestions are applicable and when they should be questioned.

Closely related to trust is perceived control. Users are more likely to engage with a system when they feel that they understand how their actions influence system outcomes and that they retain agency over final decisions. Systems that present unexplained or unclear conclusions may undermine this sense of control, even when their outputs are correct [16].

In educational planning contexts, perceived control is particularly important, as students must feel that they can reason about and influence their academic progression. Explainable feedback thus serves not only a cognitive function but also a motivational and psychological one, supporting autonomy, confidence, and sustained engagement.

2.4.5 Implications for this thesis

The design principles outlined in this section directly inform the system developed in this thesis. By supporting both minimal and detailed explanations, the system seeks to balance cognitive efficiency with transparency and learning. By framing explanations around user-relevant concepts rather than internal system mechanics, it aims to support understanding, trust, and perceived control.

These principles also motivate the evaluation approach adopted in this work, in which user understanding, trust, and perceived control are treated as central outcome variables. In this way, explainability is positioned not merely as a technical feature, but as a fundamental component of effective decision support.

3 System Design and Implementation

This chapter describes how the theoretical foundations presented in Chapter 2 are operationalized in the design and implementation of a working decision support system. The system is explicitly developed as a research artifact whose properties can be evaluated in relation to the stated research questions. The chapter, therefore, focuses on design intent, architectural structure, and reasoning mechanisms, and explains how these choices support correctness, explainability, user trust, and comparative evaluation.

3.1 Design goals and methodological approach

The design of the system is guided by the overarching research problem of how rule-based decision support for academic progression can be presented in a way that is both correct and understandable, and that fosters user trust. The central design challenge is not merely to compute whether degree requirements are fulfilled, but to communicate this assessment in a form that students can interpret, question, and rely upon when making planning decisions.

To address this challenge, the system is designed according to a set of explicit design goals that shape both its technical architecture and its user-facing interaction. These goals are derived directly from the research questions and from the conceptual foundations discussed in the background chapter.

First, **Correctness** is treated as a non-negotiable design goal due to the formal and institutional nature of degree requirements. Academic rules define explicit conditions for eligibility, and therefore, incorrect or misleading feedback may have serious consequences for students' planning decisions, such as delayed graduation or inappropriate course selection. For this reason, the system prioritizes formal correctness in its reasoning, ensuring that conclusions about requirement fulfillment follow directly from well-defined and inspectable rules.

This goal directly addresses the first sub-question of the study, which concerns the rules and logical structures required to accurately map student course data against formal degree requirements. Correctness is operationalized through the use of a declarative reasoning framework in which requirements are encoded explicitly rather than being embedded implicitly in procedural code.

While **Explainability** is treated as a first-class design concern rather than as a secondary visualization problem. While many systems are capable of computing compliance with formal requirements, fewer are designed to explain why a particular conclusion has been reached. In this system, explainability is supported by structuring the reasoning process in a way that produces intermediate results suitable for explanation.

This design goal maps directly to the research questions concerning how feedback should

be presented and how different explanation styles influence understanding. By designing the reasoning engine to expose unmet requirements, violated constraints, and alternative possibilities, the system enables explanations that reflect the underlying logic rather than obscuring it behind simplified outcomes.

Meanwhile, **user trust** is not assumed to arise automatically from correctness alone. Instead, the system is designed to support trust by making its reasoning inspectable and its conclusions open to exploration. Rather than presenting outcomes as authoritative verdicts, the system frames them as reasoned assessments that users can examine at different levels of detail.

The goal here is closely tied to the research question concerning the relationship between explainability, trust, and perceived control. By allowing users to see not only what the system concludes but also why, the design aims to support a sense of agency and reduce the perception of the system as an opaque authority.

The system explicitly supports the generation and presentation of multiple valid study trajectories. This design choice reflects the reality that academic planning rarely has a single optimal solution and that students may prioritize different trade-offs, such as workload distribution, course focus, or scheduling constraints.

Supporting multiple solutions avoids framing the system as prescribing a single "correct" path and instead positions it as a decision support tool. This directly supports the main research question by emphasizing understanding and informed choice rather than automation.

Taken together, these design goals define a methodological approach in which formal reasoning, explanation, and interaction are co-designed. The system is constructed not only to function correctly, but to serve as a research instrument for studying how different forms of feedback influence users' understanding, trust, and perceived control. The following sections describe how these goals are operationalized through the system's architecture, reasoning mechanisms, explanation generation, and user interface design.

3.2 Design grounding through expert interview

To ground the system design in actual academic advising practice, an exploratory interview was conducted with a study counsellor responsible for the Master of Science Programme in Interaction Technology and Design. The purpose of this interview was not to evaluate an existing system or to collect empirical results, but to inform design decisions by gaining insight into how degree requirements are interpreted in practice, what types of questions students commonly raise, and where current information systems fall short in supporting independent understanding of study progression.

The interview discussed the counsellor's routine work processes, the types of materials usually used when advising students, and the situations where students typically experience uncertainty. Particular attention was given to cases where students deviate from the standard study plan in their early years, such as changing specialisations, falling behind schedule, or combining courses in non-standard ways. These cases were described as especially time-consuming, often requiring manual cross-referencing between programme regulations, course plans, and examination requirements.

A frequently discussed subject in the interview was that the early stages of the programme

are relatively straightforward for most students, as recommended course sequences are clearly defined. However, as students progress into the later stages of the programme, planning becomes more individual and complex. At this point, students are expected to independently assemble a coherent study plan that satisfies multiple overlapping requirements, including subject areas, academic level, and eligibility requirements. Determining whether these requirements are fulfilled was described as requiring interpretation rather than simple credit counting.

The counsellor emphasized that requirements related to subject areas and advanced-level studies are particularly difficult for students to assess on their own. Although relevant information is formally available through programme documentation and course plans, it is distributed across multiple sources and expressed using terminology that is not easily mapped to individual study records and is not clear enough to be applied directly in some cases. As a result, students frequently seek confirmation about whether they meet specific requirements or warnings about potential deficiencies that may affect eligibility for future courses or thesis work. The interview also highlighted that existing administrative systems primarily support record-keeping rather than reasoning about degree progression. While such systems accurately display completed and registered courses, they do not explain how this information relates to programme-specific rules. Consequently, a significant portion of advising work involves manually mapping a student's course history onto degree requirements and explaining the reasoning behind the assessment.

These insights directly informed several design choices in the proposed system. First, they reinforced the importance of correctness with respect to formal degree rules, reflecting the potential consequences of incorrect guidance. Second, they motivated the focus on explanation as a central system capability rather than a secondary presentation feature. Third, they supported the decision to model planning as a problem with multiple valid solutions, reflecting the fact that students can reach the same academic goals through different combinations of courses.

The interview thus served as a qualitative basis for the system's design goals and interaction concept. The system is therefore intentionally designed as a complementary decision support tool, rather than as a replacement for full academic counseling. Its purpose is to make formal rules visible, link them to the individual's study data, suggest multiple plans, and present the information in a way that encourages understanding and informed discussions – both when used independently and in preparation for meetings with academic counselors.

3.3 System overview and architecture

To operationalize the design goals outlined in the previous section, the system is organized as a modular decision support architecture in which reasoning, explanation, and presentation are clearly separated. This separation of concerns is a deliberate design choice intended to support correctness, explainability, and experimental flexibility.

At a high level, the system consists of three main components:

1. a backend responsible for data preparation and solver orchestration,
2. a reasoning layer implemented using an Answer Set Programming (ASP) solver, and

3. a frontend responsible for presenting results and explanations to users.

Figure 1 provides a high-level overview of the system architecture and illustrates how information flows between these components, from student and program data to generated study trajectories and user-facing explanatory feedback.

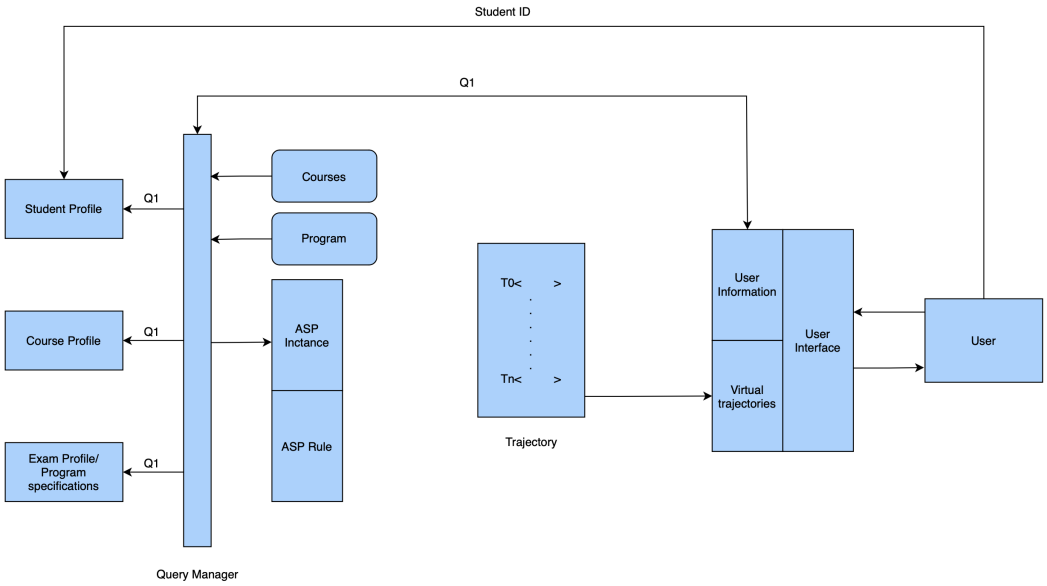


Figure 1: High-level system architecture of the decision support system. The diagram illustrates the flow from student and program data through the ASP-based reasoning engine to the generation of study trajectories and explanatory feedback presented to the user.

As shown in Figure 1, the backend aggregates information from multiple sources, including the student profile, course descriptions, and program-specific degree and thesis requirements. These data sources are accessed and coordinated through a query management component that prepares a unified representation suitable for formal reasoning. The backend does not perform interpretation or explanation; its role is limited to managing data flow, normalizing inputs, and invoking the reasoning engine.

The reasoning layer constitutes the conceptual core of the system. Here, degree requirements, prerequisites, and planning objectives are evaluated using a declarative rule model expressed in ASP. Based on the provided input, the solver generates one or more candidate study trajectories, each representing a possible progression from the student's current academic state toward fulfillment of degree requirements. In addition to suggested trajectories, the reasoning output includes structured information about unmet requirements, violated constraints, and optimization costs. These reasoning outputs form the basis for all subsequent explanation generation.

The frontend is responsible for presenting the outputs of the reasoning layer to the user. Rather than embedding domain logic, it consumes structured reasoning and explanation data and renders them in a consistent interface. This includes both representations of the student's current academic situation and a set of virtual trajectories that illustrate alternative paths forward. Explanatory feedback is presented incrementally through interaction,

allowing users to explore additional detail without altering the underlying reasoning results.

A key architectural principle is the strict separation between rules, explanations, and the user interface. Rules define the logical conditions under which requirements are fulfilled or violated. Explanations are derived from the outcomes of these rules rather than being encoded within them. The user interface operates solely on explanation structures rather than raw solver output. This ensures that explanations faithfully reflect the system's reasoning and avoids post hoc interpretation at the presentation layer.

This separation also supports the research methodology of the thesis. Because reasoning and presentation are disconnected, identical logical conclusions can be exposed through the same interface while varying the amount of explanatory detail revealed to the user. This enables controlled comparison of explanation granularity during evaluation without confounding differences in underlying logic or plan content.

In this way, the system architecture does not merely support functional correctness, but actively enables the observed investigation of how different forms of feedback presentation influence users' understanding, trust, and perceived control, as addressed in the evaluation phase of the thesis.

3.4 ASP-based reasoning engine

The ASP-based reasoning engine constitutes the conceptual core of the system's planning capabilities. To support reasoning about academic progression over time, the system models course planning as a dynamic problem involving states, actions, and alternative trajectories. This planning perspective provides the foundation for the declarative reasoning and optimization mechanisms described in the following sections.

The choice of Answer Set Programming (ASP) reflects the need to reason over complex constraints, combinatorial decision spaces, and multiple competing objectives in a principled and transparent manner. Rather than defining a fixed sequence of decisions, the reasoning engine evaluates entire study trajectories relative to formal academic requirements and planning goals.

3.4.1 Planning as a state-based reasoning problem

In order to support planning, explanation, and adaptation, the system conceptualizes student progression as a dynamic process rather than as a purely static evaluation of degree requirements. This perspective is inspired by classical planning theory, in which reasoning is performed over states, actions, and their effects, allowing both current situations and possible future developments to be considered.

In this context, a state represents the academic situation of a student at a given point in time. A state is defined by information available to the system, including the degree program the student is enrolled in, the set of courses that have been successfully completed, and the courses for which the student is currently registered. This information forms the initial state of the planning problem and reflects the student's present progression toward degree completion.

Actions correspond to academic activities that may change the student's state. In the sys-

tem, actions primarily represent the successful completion of courses. Each such action has well-defined effects: completing a course adds credits, may satisfy prerequisites for other courses, and contributes toward fulfilling degree requirements. Whether an action is applicable depends on the current state, for example, whether prerequisite conditions are met.

Rather than procedurally simulating state transitions, the system represents states, actions, and their effects declaratively using logical rules. Possible transitions are reasoned about implicitly by the ASP solver through the interaction of these rules and constraints. This allows the system to evaluate entire study trajectories as coherent wholes, rather than constructing them step by step through explicit control flow.

A trajectory can thus be understood as one possible sequence of academic states resulting from the application of allowable actions over time. Importantly, multiple distinguishable trajectories may exist for the same student. Different combinations of courses, taken in different orders, may all lead toward fulfillment of degree requirements while emphasizing different academic areas or trade-offs. This reflects the intrinsic flexibility of higher education curricula, which also motivates the need for reasoning about alternative trajectories rather than defining a single path.

This planning-based formulation also supports the ability to change. If a student does not complete a planned course or if new information becomes available regarding courses or exam requirements, the current state will be updated accordingly. The system can then re-evaluate the situation and generate new trajectories that reflect the updated state. This ability to re-plan is essential in real educational settings, where student progression is rarely linear or fully predictable, which is one of the main reasons to implement this system from the beginning.

While this state-based planning perspective defines how academic progression is conceptualized, it does not prescribe how such reasoning should be implemented. The following sections describe how this model is realized using declarative rules, constraints, and optimization in Answer Set Programming.

3.4.2 Declarative modeling of academic rules

The state-based planning model described above is implemented using a declarative rule-based representation. In this approach, academic regulations are encoded as logical rules that specify the conditions under which actions are possible and the effects those actions have on the student's progression. States and actions are not represented explicitly as mutable data structures, but implicitly through facts and rules that describe which course completions are admissible and what consequences they entail.

ASP is particularly suitable for modelling academic regulations because it allows rules to be expressed declaratively, closely mirroring the structure of degree requirements as they appear in official documentation. In a declarative model, rules specify what must be held, not how to execute it. This distinction supports both correctness and explainability by separating domain knowledge from procedural control logic. Within the system, different categories of rules capture distinct aspects of academic progression. Prerequisite rules define dependency relations between courses. Credit accumulation rules express minimum requirements for degree areas and total credits. Additional rules capture classification constraints such as subject area membership or course level. Together, these rule categories

form a logical model that reflects the institutional definition of degree compliance. By separating rule categories conceptually, the system enables clear reasoning about why a particular plan is feasible or infeasible and which requirements contribute to violations. This structure also supports maintainability, as changes to degree regulations can be reflected by modifying rule definitions without altering the reasoning mechanism itself.

The complete set of declarative rules implementing these constraints and requirements is provided in Appendix A.1. The appendix includes the full Answer Set Programming specification used by the reasoning engine, allowing the formal realization of prerequisite handling, credit accumulation, and requirement evaluation to be inspected in detail. The presentation in this section focuses on the conceptual structure of the rules rather than their concrete syntax.

3.4.3 Constraints and optimization

Given this declarative representation of states and actions, the system distinguishes between conditions that must always hold and objectives that express progress toward degree completion. This distinction is realized through the use of hard constraints and optimization criteria.

Hard constraints represent conditions that must never be violated, such as prerequisite ordering or course availability. Any plan violating such constraints is considered invalid and excluded from the solution space. These constraints ensure that all generated trajectories respect fundamental academic rules.

In contrast, degree requirements such as minimum credits in specific areas are treated as optimization objectives rather than absolute constraints. This allows the system to generate meaningful partial solutions even when full compliance is not yet achievable. Instead of failing outright, the system can identify plans that minimize unmet requirements and maximize overall progress. The concrete optimization criteria and their prioritization are formally specified in the ASP program included in Appendix A.1.

This approach reflects real-world academic planning, where students often seek guidance every term, long before they are eligible for graduation. Encoding degree fulfillment as a graded objective rather than a binary condition supports incremental feedback and comparison between alternative trajectories.

3.4.4 Multiple solutions and optimal models

Because the planning problem is underdetermined, multiple solutions may satisfy the constraints equally well or differ only marginally in quality. The ASP solver is therefore configured to identify optimal models according to defined cost functions that capture remaining requirement violations and progression deficits.

The system does not assume the existence of a single “correct” plan. Instead, it supports the generation of multiple optimal or near-optimal trajectories, each representing a valid interpretation of the constraints. This design choice aligns with the decision support perspective adopted in this thesis, in which the system assists users by presenting alternatives rather than prescribing outcomes.

Importantly, the solver’s cost structure is exposed to later stages of the system, enabling

trajectories to be compared, ranked, and explained in a principled manner. Optimization thus functions as an integral part of both reasoning and explanation, rather than as a hidden internal mechanism.

3.5 Explanation generation

A central design objective of the system is to ensure that formally correct reasoning about degree requirements can be communicated to users in a manner that is understandable, trustworthy, and supportive of informed decision-making. While the ASP-based reasoning engine described in the previous section determines which study trajectories are feasible or optimal, explanation generation addresses the complementary question of how these conclusions should be presented to users. Therefore, explanation generation is treated as an integral part of the system design rather than a secondary presentation layer.

3.5.1 Deriving Explanations from Formal Reasoning

Explanations in the system are not predefined for individual courses or requirements. Instead, they are derived directly from the outcomes of the reasoning process. The ASP solver delivers structured information about unmet requirements, violated constraints, and infeasible course choices, in addition to suggested course selections. These reasoning outputs form the basis for all user-facing explanations.

This design choice ensures a tight connection between correctness and explainability. Because explanations are computed from the same logical structures that determine feasibility and optimization, the system avoids distinctions in which feedback may appear reasonable but is not formally justified. As a result, explanations are not post hoc interpretations layered on top of system output, but faithful representations of the reasoning that led to a given conclusion. Therefore, by avoiding hard-coded feedback, the system also remains robust to changes in degree requirements or changes in the reasoning logic. Updates to the underlying rules are automatically reflected in the generated explanations, supporting maintainability and consistency over time.

3.5.2 Explanation structures

The outputs produced by the solver are not directly suitable for presentation to end users, even while the reasoning engine operates over logical rules and constraints. To address this, the system introduces an intermediate layer of explanation structures that organize reasoning outcomes into semantically meaningful categories. The system distinguishes between different types of explanatory information corresponding to distinct reasoning outcomes. These include, for example, unmet degree area requirements, low total credits, missing prerequisites, and courses that cannot be scheduled within the given constraints for some reason. Each category reflects a specific logical conclusion drawn by the solver, such as the failure to satisfy a required credit threshold or the impossibility of completing a prerequisite chain within the available planning horizon or due to course offerings.

The system maps these conclusions into structured explanation objects that preserve semantic meaning while remaining accessible to non-expert users, rather than presenting them as

raw logical statements. This abstraction layer enables the system to decouple explanation content from presentation format, making it possible to support multiple explanation styles without altering the underlying reasoning model.

Rather than presenting these conclusions as raw logical statements, the system translates them into structured explanation objects that preserve semantic meaning while remaining understandable to non-expert users. This abstraction layer separates the explanation content from the presentation form and enables support for multiple explanation styles without the need to change the underlying reasoning model.

3.5.3 Minimal and detailed explanations

To enable systematic investigation of how explanation level influences user understanding and trust, the system provides two distinct explanation modes: a minimal mode and a detailed mode. These modes are intended to function as controlled conditions for observation and compassion.

The minimal explanation mode presents high-level feedback, such as whether degree requirements are fulfilled or which requirement categories lack credits. This mode emphasizes outcome-oriented information and reduces the amount of explanatory detail shown to the user.

The detailed explanation mode reveals additional causal information for these outcomes, such as missing prerequisite courses, credit distributions across degree areas, reasons for course infeasibility, or specifies the covered area for every course in the plan. Importantly, this information is derived from the same underlying reasoning structures as the minimal explanations and differs only in the level of detail presented.

By keeping the underlying reasoning and plan content constant while varying explanation granularity, the system makes it possible to examine how different forms of feedback presentation affect users' understanding, trust, and perceived control. The two modes thus function as experimental instruments rather than as prescriptive recommendations for interface design.

3.5.4 Implications for trust and perceived control

The separation between formal reasoning, explanation generation, and presentation is a conscious design decision intended to support observed investigation of user trust, understanding and perceived control. Because explanations are derived rather than scripted, the system ensures that feedback is grounded in the same formal logic that governs its reasoning outcomes.

The availability of both minimal and detailed explanation modes allows users to access different levels of information, making it possible to study how explanation influences users' ability to interpret system feedback, verify conclusions, and reason about alternative trajectories. Rather than assuming that increased detail necessarily leads to greater trust or understanding, the system is designed to expose this relationship to observed evaluation.

In this way, explanation generation functions not only as a communication mechanism but also as an experimental bridge between formal reasoning and user perception. This design directly operationalizes the research questions concerning how explainability and presenta-

tion style affect students' understanding, trust, and perceived control.

3.6 User interface design

The user interface is designed to operationalize the system's focus on explainability, transparency and trust by presenting reasoning results to users with the ability to control the feedback. Rather than optimizing for efficiency or personalization, the interface is intentionally structured to support observed investigation of how different forms of feedback presentation influence users' understanding, trust, and perceived control and transparency for the suggested study plans.

Building on the explanation generation mechanisms described in the previous section, the interface serves as the final layer. Its primary role is therefore not to introduce new reasoning or interpretation, but to expose existing explanation structures in a consistent and comparable manner.

Screenshots documenting the implemented user interface consisting of a start page and a study progression view with minimal and detailed explanations are provided in Appendix A.3

3.6.1 Design context and alignment with Ladok

The system developed is designed with the explicit intention that its functionality form part of the Ladok environment from a user perspective. Therefore, the user interface is designed to align with Ladok's existing visual and interaction system. This alignment reflects both the institutional context of the thesis and the intended use case of the proposed functionality: to support students within the system they already use to access information about their courses, credits, and academic status.

At the same time, the implementation presented in this thesis is a standalone prototype developed for research and evaluation purposes. It is not technically integrated with Ladok and does not interact with live student data. Instead, it operates on representative data and degree definitions in order to explore design alternatives and study how different forms of explanatory feedback are perceived by users.

Within this context, the prototype is presented under the name "Min studieprogression", reflecting how such functionality could plausibly be introduced as a feature in Ladok. The interface includes a progression-oriented view that summarizes the student's current academic status, as well as a separate view that presents degree and thesis requirements specific to the student's enrolled program. This structure mirrors existing Ladok views while extending them with decision support and explanatory capabilities.

By designing the system for the Ladok context while implementing it as a prototype, the thesis balances ecological validity with methodological control and data safety. This approach allows the evaluation to focus on explanation design and user perception without implying deployment, institutional endorsement, or operational integration.

3.6.2 Planning view and progression context

The planning view presents suggested study plans grouped by academic term and total credits, making the structure of the progression explicit. This representation reflects how students typically reason about course planning, where decisions are organized around semesters and workload distribution rather than abstract rule satisfaction alone.

While presenting plans in a term-based structure provides a stable contextual reference for the user to interpret feedback about feasibility and requirement fulfillment, it also enables users to relate abstract constraints, such as prerequisites or credit limits, to concrete planning decisions, such as when a course can be taken or how different choices affect future options.

3.6.3 Reasoning panel and separation of concerns

Progression feedback is displayed in a dedicated reasoning panel, separate from the planning view itself. This design choice reflects the architectural principle that reasoning and presentation should remain separated while still being connected through interaction.

By separating the representation of a suggested plan from explanations about its feasibility or requirement satisfaction, the interface avoids mixing what is planned with why it is or is not valid. This separation is particularly important for evaluation, as it allows users to focus on explanatory feedback without changes to the underlying plan content.

3.6.4 Minimal and detailed explanation modes as interface conditions

The user interface displays the minimal and detailed explanation modes described in Section 3.4 as alternative interface configurations. These modes do not differ in underlying reasoning or explanation content, but solely in how much of the available explanatory information is made visible to the user at a given time.

From an interface perspective, the distinction between modes concerns presentation and interaction rather than explanation logic. The minimal mode restricts visible feedback to high-level outcomes, while the detailed mode allows users to inspect additional explanatory elements derived from the same explanation structures. No new conclusions are introduced when clicking to view more feedback and switch between modes; the difference lies exclusively in the degree of informational exposure.

Crucially, the interface implements explanation granularity through progressive disclosure within a single, consistent view. All users initially encounter the same minimal feedback, and additional explanatory detail is revealed incrementally through interaction, such as selecting “view more” options. Layout, terminology, and overall structure remain constant as explanations are expanded, ensuring that changes in visible information are attributable solely to explanation depth rather than to interface variation.

By implementing minimal and detailed explanations as interface conditions rather than distinct system behaviors, the user interface enables controlled comparison of explanation presentation styles. This directly supports the evaluation strategy of the thesis by allowing explanation granularity to be treated as an experimental variable.

3.6.5 Consistency and evaluation support

To ensure comparability during user evaluation, the interface prioritizes consistency across all interactions. A single, stable layout is used throughout, with consistent terminology and interaction patterns. Users initially encounter minimal feedback, and additional explanatory detail is revealed incrementally through interaction, while the overall visual and structural context remains unchanged.

This design reduces the risk that observed differences in user responses can be attributed to unrelated interface factors such as layout changes or navigation complexity. Instead, it allows the evaluation to focus on how users engage with increasing levels of explanatory detail within a consistent interface context, directly supporting the research questions concerning understanding, trust, and perceived control.

3.7 Functional scope and system guarantees

The system developed in this thesis is designed as a decision support tool for academic study planning rather than as an automated planning or decision-making system. Its functional scope is therefore deliberately constrained. This section clarifies the guarantees the system is designed to provide, the aspects that lie outside its intended scope, and how these boundaries support the overall research goals.

A central design guarantee of the system is formal correctness with respect to explicitly encoded degree requirements. All study plans generated by the reasoning engine are constructed to respect formal prerequisite relations, credit constraints, degree and area requirements, and temporal ordering of courses. By relying on a rule-based reasoning framework, the system is designed to ensure that proposed study trajectories do not violate known institutional rules. Courses are only included when prerequisite conditions can be satisfied, credit accumulation is tracked consistently, and degree requirements are evaluated according to their formal definitions. This guarantee is non-negotiable, as incorrect feedback in the context of academic planning may lead to delayed graduation, invalid course choices, or misplaced trust in system output.

In addition to correctness, the system is designed to guarantee traceability of reasoning outcomes. Rather than producing a single unclear result, the reasoning process exposes unmet requirements, violated constraints, and optimization trade-offs as explicit outputs. This enables the system to explain not only what the current academic status is, but why certain requirements are considered fulfilled or unmet. The guarantees provided by the system, therefore, extend beyond computational validity to include the availability of structured information that supports explanation and inspection.

The system also guarantees support for multiple valid study trajectories. Academic degree structures often permit several alternative paths toward completion, particularly in later stages of study involving specialization and elective choices. Instead of enforcing a single "best" plan through hard constraints, the system models study planning as an optimization problem. This allows multiple solutions to coexist, each satisfying formal constraints while differing in how remaining requirements are addressed. Therefore, it could be claimed that this design reflects the inherent flexibility of academic programs and avoids presenting planning as a deterministic process with a single correct outcome.

At the same time, the system does not aim to guarantee that generated study plans are optimal with respect to individual student preferences, workload balance, career goals, or personal circumstances. Such factors are inherently subjective and cannot be fully captured through formal rules alone. The system, therefore, refrains from making prescriptive decisions on behalf of students. Instead, it provides formally valid alternatives and structured feedback that users can interpret and evaluate in light of their own goals and constraints.

This distinction between guaranteed formal correctness and non-guaranteed subjective optimality is central to the system's function as a decision support tool. By ensuring that all suggested plans comply with formal rules while leaving the evaluative judgments to the user, the system supports informed decision-making without undermining user autonomy. This design position directly reflects the research focus of the thesis, which prioritizes user understanding, trust, and perceived control over automated decision-making authority.

In summary, the system is designed to ensure that its feedback is formally correct, logically transparent, and grounded in explicit degree requirements. It does not guarantee that its suggestions represent the best possible plan for every student, nor does it aim to replace human advising. By clearly defining both its guarantees and its limitations, the system provides a stable foundation for explainable feedback and meaningful interaction, while appropriately constraining its scope to support the design objectives and the subsequent evaluation of user understanding and trust.

4 Evaluation

4.1 Evaluation goals and research questions

The purpose of the evaluation is to investigate how users perceive and interpret feedback produced by the proposed rule-based decision support system for study progression. In contrast to evaluations that focus on system performance or optimization accuracy, this evaluation is concerned with users' experienced understanding, trust in system feedback, and perceived control when interacting with the system.

The evaluation is designed to address the main research question of the thesis: *How can a rule-based decision support system for course progression be designed so that students understand and trust its feedback regarding their fulfillment of degree requirements?*

In particular, the evaluation focuses on the following aspects derived from the sub-questions:

- how students understand the relationship between their course history and formal degree requirements when this relationship is explicitly explained,
- how different levels of explanation detail influence perceived clarity and usefulness of feedback,
- how system transparency affects students' trust in the system and their perceived ability to reason about their own academic progression.

Because the system is intended as a decision support tool rather than an automated decision-making system, the evaluation does not aim to assess whether users follow the system's suggestions or whether the generated study plans are objectively optimal. Instead, the evaluation examines whether the system succeeds in making formal degree rules understandable, interpretable, and actionable from the user's perspective.

The evaluation is exploratory and formative in nature. Its goal is not to produce statistically generalizable results, but to gain insight into how the proposed design choices function in practice and how they are experienced by intended users in a realistic academic context. This approach is appropriate given the prototype status of the system and the focus on explanation, trust, and perceived control as subjective and context-dependent phenomena.

4.2 Study design and procedure

The evaluation consists of two complementary parts:
(1) a pilot study involving a study counsellor, and
(2) a user study involving students enrolled in a relevant degree programme.
Together, these parts provide both expert-informed assurance of the system's relevance and user-centered insight into how the system's feedback is perceived in practice.

The system was evaluated as a standalone prototype designed for the Ladok context. During the study, participants were informed that the system should be understood as a conceptual extension of Ladok, presented as two additional pages in the Ladok menu. This framing was intended to situate the interaction in a familiar institutional setting while avoiding any implication of technical integration with existing administrative systems. The prototype consisted of two main interface views: a study progression overview and a requirements-focused view presenting degree and thesis eligibility conditions. The interface supported two explanation modes, in which feedback was initially presented in a minimal form and could be expanded to reveal more detailed explanations of unmet requirements, course dependencies, and alternative study paths. The interface as presented to participants during the evaluation is documented through screenshots in Appendix A.3.

Overall procedure

For the student user study, the evaluation followed a structured but low-pressure procedure designed to encourage exploration rather than task optimization:

1. Participants received a brief introduction to the purpose of the study and the role of the system as a decision support tool for study progression.
2. A study scenario was presented, in which participants were asked to assume the role of a student enrolled in the Master of Science Programme in Interaction Technology and Design and currently in the second or third year of the programme.
3. Participants were introduced to the prototype and informed that it represented a potential Ladok feature titled “Min studieprogression”, including a progression overview and a page showing programme-specific degree and thesis requirements.
4. Participants were given time to freely explore the system, inspect their academic progression, examine alternative study trajectories, and interact with explanatory feedback. The system initially presented high-level feedback, with additional explanations revealed incrementally through interaction.
5. After completing the exploration, participants were asked to fill out a questionnaire reflecting on their experience of the system.

No time limits were imposed, and participants were encouraged to focus on understanding the feedback rather than completing predefined tasks. This approach reflects the exploratory nature of academic planning and aligns with the research focus on understanding and interpretation rather than efficiency or performance.

Rationale for evaluation approach

The evaluation prioritizes self-reported user experience, gathered through a questionnaire, rather than objective measures of task performance. This methodological choice follows from the nature of the research questions, which focus on perceived understanding, trust, and control—constructs that cannot be reliably derived from interaction logs or correctness metrics alone.

By combining an expert-informed pilot study with a user study situated in a realistic academic context, the evaluation seeks to capture a nuanced understanding of how the system’s design decisions operate in practice and how they are perceived by the intended user group.

4.3 Pilot study with study counsellor

Prior to the student user study, a pilot evaluation was conducted with a study counsellor responsible for the Master of Science Programme in Interaction Technology and Design. The purpose of this pilot was not to assess usability, interface aesthetics, or user satisfaction, but to perform a formative expert evaluation of the system's reasoning outcomes and explanation logic from an academic advising perspective.

More specifically, the pilot aimed to examine whether the system's conclusions about study progression were correct and professionally reasonable, whether the generated explanations reflected how a study counsellor would reason about degree requirements, and whether the system expressed its feedback in a way that would not risk misleading students. The pilot thus served as a qualitative validation of the system's rule-based reasoning and explanation design before conducting the student-facing evaluation.

Procedure and focus

The pilot followed a structured, scenario-based format. After a brief introduction to the system and its purpose as a rule-based decision support tool, the counsellor was presented with a predefined student case. The case was selected to maximize explanatory content and included a situation in which degree requirements were not yet fulfilled.

Using this scenario, the counsellor reviewed the system's assessment of the student's progression and interacted with the interface to inspect different levels of explanatory detail. The review focused on four main aspects:

1. **Correctness of conclusions**, assessing whether the system's judgments aligned with how the counsellor would evaluate the student's situation based on formal degree requirements.
2. **Professional plausibility of explanations**, examining whether the explanations correspond to how an expert would typically explain unmet requirements to a student.
3. **Risk of misinterpretation**, identifying cases where system feedback might be misunderstood or taken as more authoritative than intended.
4. **Explanation granularity**, comparing minimal feedback with more detailed explanations, and discussing whether the level of detail was appropriate, insufficient, or excessive from an advising perspective.

The counsellor was encouraged to reflect aloud and to comment on both strengths and potential risks associated with the system's feedback. Notes were taken during the session, and no quantitative measures were collected.

Outcomes and role in the study

The pilot did not aim to produce generalizable findings, but to inform and refine the system and the subsequent user study. Feedback from the counsellor confirmed the relevance of focusing on explanation as a core system capability and highlighted the importance of carefully phrased feedback, particularly in cases where requirements were not fulfilled.

The pilot also reinforced the distinction between providing informative feedback and replacing academic advising. While the system was seen as potentially valuable for helping

students understand formal requirements and prepare for advising meetings, the counsellor emphasized that personal context and individual circumstances remain essential aspects of study guidance.

Insights from the pilot informed minor refinements to terminology, explanation phrasing, and the framing of requirement fulfillment, and helped ensure that the system's feedback aligned with professional advising practices. The pilot thus functioned as expert grounding for the evaluation design, supporting the validity of the subsequent student user study.

In addition to feedback on correctness and explanation logic, the counsellor expressed a positive assessment of the system's potential role in academic advising. In particular, the system was seen as a valuable preparatory tool that could help students gain an initial understanding of degree requirements and possible study paths before seeking individual guidance.

The counsellor indicated that, in its current conceptual form, the system will be suited to be recommended to students as a first step when they are uncertain about their progression or future course choices. From an advising perspective, this was described as potentially beneficial for both students and counsellors, as it could help students arrive at advising meetings better prepared, with more concrete questions and a clearer understanding of their situation.

This feedback further supports the positioning of the system as a complementary decision support tool rather than a replacement for academic advising. The system's role is not to provide personalized judgments, but to make formal requirements explicit and support more informed and efficient dialogue between students and study counsellors.

4.4 User study with students

Following the pilot evaluation with a study counsellor, a user study was conducted with students in order to investigate how the proposed system is perceived by its intended users. The purpose of this study was to examine students' understanding of system feedback, their trust in the system's conclusions, and their perceived control over academic planning when interacting with the prototype.

The user study was exploratory in nature and focused on users' subjective experiences rather than on objective task performance. This approach is appropriate given the research focus on explainability, transparency, and trust, which are inherently experiential and context-dependent.

4.4.1 Participants and context

The study involved students enrolled in the Master of Science Programme in Interaction Technology and Design. Participants were distributed among students from the third, fourth, and fifth years of the programme, a stage at which study planning typically becomes more complex due to choices, specialization decisions, and eligibility requirements for thesis work.

All participants were familiar with Ladok as the primary system used to access information about courses, credits, and academic results. This familiarity was important for situating

the prototype in a realistic institutional context and for minimizing learning effects related to interface conventions rather than system functionality.

The number of participants was limited to 14, and the study does not aim to produce statistically generalizable results. Instead, it seeks to provide qualitative and descriptive insight into how students experience the system and its explanations in practice.

4.4.2 Study scenario and tasks

Participants were introduced to the system through a designed but realistic study scenario. They were asked to assume the role of a student enrolled in the Master of Science Programme in Interaction Technology and Design and currently in the second or third year of the programme. The scenario was intentionly designed to reflect a typical situation where it is time for a student to plan their own individual plans, while the student is uncertain about their progression, degree requirements, or eligibility for future studies.

The system was presented as a prototype designed for the Ladok context. Participants were informed that the interface represented a potential extension of Ladok, accessible through two additional pages in the Ladok menu. One page provided an overview of the student's study progression, while the other presented programme-specific degree and thesis requirements.

Participants were explicitly informed that the system functioned as a decision support tool rather than an authoritative decision-maker. Its purpose was described as helping students understand formal degree requirements and explore alternative study trajectories, not as replacing academic advising or making final decisions on their behalf.

4.4.3 Interaction and tasks

Participants were given time to freely explore the system without strict task constraints. Rather than completing predefined tasks, they were encouraged to navigate the interface as they normally would when trying to understand their academic status. This included examining their current progression, inspecting unmet degree requirements, exploring alternative study trajectories, and interacting with explanatory feedback.

The system initially presented high-level feedback regarding requirement fulfillment and progression status. Additional explanatory detail was revealed incrementally through interaction, allowing participants to access more detailed explanations when desired. This interaction pattern reflects the progressive disclosure design described in Chapter 3 and enabled participants to engage with different levels of explanation during the study.

No time limits were imposed, and participants were encouraged to ask clarifying questions about the study setup if needed. However, no guidance was provided on how to interpret the system's feedback, in order to observe how participants made sense of the information independently.

4.4.4 Questionnaire design

After completing the exploration phase, participants were asked to complete a questionnaire reflecting on their experience of the system. The questionnaire consisted of a combination

of Likert-scale items and open-ended questions.

The questionnaire was designed to capture participants' perceptions across several aspects relevant to the research questions, including:

- perceived understanding of degree requirements and study progression,
- clarity and usefulness of explanations,
- trust in the system's feedback,
- perceived control and ability to reason about future study choices.

Likert-scale questions provided a structured way to assess overall tendencies in participant responses, while open-ended questions allowed participants to elaborate on their reasoning, express uncertainties, and suggest improvements. This combination enabled both explanatory analysis and qualitative interpretation of user experiences.

The questionnaire focused on self-reported perceptions rather than objective correctness. This choice reflects the research aim of understanding how explanation design influences users' sense of understanding and trust, rather than measuring whether users reach particular decisions or outcomes.

4.5 Results

This section presents the results of the student user study. The results are organized according to the main evaluation dimensions derived from the research questions: perceived understanding, explainability and information level, trust in system feedback, perceived control, and experiences related to explanation depth. The focus is on describing participants' responses rather than interpreting their implications, which is addressed in the subsequent discussion chapter.

4.5.1 Perceived understanding of study progression

Participants generally reported that the system supported their understanding of their academic progression and degree requirements. This was reflected in open-ended responses to Q7, where most participants described the system's assessment as helpful for forming an overview of their current study situation. In particular, participants highlighted that the progression overview provided a clearer picture of unmet requirements and remaining obligations.

Several participants noted that seeing requirements explicitly connected to their completed and planned courses made it easier to understand why certain requirements were considered fulfilled or unmet. This was reflected both in Likert-scale responses to Q8, where a majority of participants rated the system's explanations as clear or very clear (Table 1), and in open-ended responses to Q9 describing how the mapping between courses and requirements supported comprehension. One participant remarked that *"it became much clearer why certain courses did not count once I could see how the requirements were broken down"* (P4).

Table 1 Did you feel that the system clearly explained why the degree requirements were met or not?? (Q8, n = 14).

Response option	n	%
Not at all	0	0%
Quite unclear	0	0%
Neither clear nor unclear	0	0%
Quite clear	9	64,3%
Very clear	5	35,7%

This explicit connection between course data and formal requirements was frequently described as an improvement compared to existing administrative systems, where course records and degree requirements are typically presented separately. Participants indicated in Q9 that the integrated view reduced the need to manually cross-reference information across multiple sources.

At the same time, some participants reported initial uncertainty when first interacting with the system in the minimal explanation view. This uncertainty was primarily related to understanding how requirement categories were defined and grouped, as described in responses to Q11. However, responses to Q12 suggest that this uncertainty was often reduced as participants explored additional explanatory information. Several participants described how accessing more detailed explanations helped clarify requirement structures and credit distribution (P7, P11).

Overall, the results indicate that the system provided a clearer mental model of degree progression for most participants. The findings also suggest that explanation clarity and terminology play an important role, and that presenting more detailed explanations earlier — or keeping the detailed view as the initial state of the system — may further improve initial understanding.

4.5.2 Explainability and information level

Participants' responses indicate that explanation played a central role in how they experienced the system. Responses to Q10 suggest that most participants perceived the balance between overview-level and detailed information as appropriate, indicating that the system generally provided a suitable amount of information for understanding study progression (Table 2).

Table 2 Perceived balance between overview-level and detailed information in the system (Q10, n = 14).

Response option	n	%
Too little information	0	0%
Slightly too little	1	7,1%
Appropriate amount	12	85.7%
Slightly too much	1	7,1%
Too much information	0	0%

At the same time, several participants reported in Q11 that minimal feedback alone was not always sufficient to fully understand their situation, particularly in cases involving subject-

area requirements or prerequisites constraints. Where participants expressed a need for additional context in order to interpret the system's conclusions.

The ability to reveal additional explanatory detail through interaction was therefore described as valuable. In responses to Q12, participants emphasized that detailed explanations helped them understand how requirements were evaluated and why certain courses did or did not contribute toward degree fulfillment. One participant noted that *"the detailed view made it easier to see what was missing and why"* (P2).

At the same time, some participants noted that too much information presented at once could feel overwhelming. Some responses to Q11 indicated that overly detailed explanations could feel overwhelming if shown by default. In this context, progressive disclosure was frequently described as a positive design feature, as it allowed users to control how much detail they engaged with based on their immediate needs (Q12).

Overall, the results suggest that participants appreciated having both overview-level feedback and and the option to access more detailed explanations, rather than being presented with a fixed level of information. These findings highlight the importance of flexible explanation depth in supporting both efficiency and understanding.

4.5.3 Trust in the system

Trust in the system was closely tied to perceptions of correctness and transparency. Responses to Q13 indicate that most participants reported a high degree of trust in the system's assessment of study progression (Table 3). Participants frequently described the system's feedback as reliable, particularly when explanations clearly showed how conclusions were derived from formal degree requirements and course data.

Table 3 To what extent participants trusted the system's assessment of study progression (Q13, n = 14).

Response option	**n**	**%**
Not at all	0	0%
Low extent	0	0%
Moderate extent	1	7.1%
High extent	5	35.7%
Very high extent	7	57.1%

Several participants referred in Q15 to the system's alignment with Ladok as an important factor contributing to trust. Presenting the system as a potential Ladok feature appeared to reinforce the perception that the information was institutionally grounded and rule-based rather than hypothetical. One participant noted that "the fact that it looked and felt like Ladok made the information feel more trustworthy" (P2).

At the same time, responses to Q14 reveal that some participants emphasized the importance of understanding the system's limitations. While they generally trusted the system's evaluation of formal requirements, they expressed hesitation about relying on it for decisions involving personal preferences, workload balance, or long-term planning. These considerations were described as lying outside the scope of what a rule-based system could reasonably guarantee.

Overall, the results suggest that transparency and explanation were key factors influencing trust, while participants remained aware that the system should function as a decision support tool rather than an authoritative decision-maker. Trust was therefore not unconditional, but calibrated in relation to the system's clearly perceived role and limitations.

4.5.4 Perceived control and actionability

Participants' responses indicate that the system supported a sense of control over academic planning by making the requirements and the consequences of course choices more explicit. Responses to Q16 suggest that most participants felt the system provided sufficient information to understand what they could or should do next in their studies (Table 4). Several participants described how the feedback helped them identify concrete next steps, such as which types of courses were needed to fulfill specific degree requirements.

Table 4 Did you feel that the system gave you enough information to understand what you can or should do next in your studies? (Q16, n = 14).

Response option	**n**	**%**
Not at all	0	0%
To low extent	0	0%
To some extent	0	0%
To a great extent	12	85.7%
To a very high extent	2	14.3%

This sense of control was further reflected in responses to Q17, where participants reported that the system's feedback contributed to a feeling of being in control of their study progression. Participants emphasized that understanding formal constraints, such as missing requirements or prerequisite dependencies, made it easier to reason about future choices rather than feeling uncertain or blocked.

The presentation of alternative study trajectories was perceived as particularly useful by some participants, as reflected in open-ended responses to Q22. Participants described how seeing multiple possible paths toward degree completion illustrated that there is often more than one viable way to progress, which reduced anxiety associated with planning and made the system feel less prescriptive. One participant noted that *"it felt reassuring to see that there were several possible ways forward instead of just one correct plan"* (P3).

At the same time, participants emphasized that meaningful planning decisions still require personal judgment. While the system was viewed as helpful for clarifying formal requirements and preparing for decisions, it was not perceived as a replacement for individual reflection or study counselor. This perspective was expressed across responses to Q22, where participants described the system as a tool for preparation and support rather than an authority that should determine final choices.

Overall, these findings suggest that the system supported perceived control by clarifying formal constraints and possible actions, while leaving value-based decisions and personal priorities to the user.

4.5.5 Experiences of explanation: minimal vs detailed explanations

Participants' experiences highlight differences in how explanation depth was used during interaction. Responses to Q18 indicate that most participants preferred the detailed explanation version when asked which version best supported their understanding of study progression (Table 5). However, open-ended responses to Q19 reveal variation in how participants engaged with explanation depth during interaction.

Table 5 Participant preferences regarding explanation depth and trust (Q18 and Q20, n = 14).

Question	Minimal	Detailed	No difference
Q18: Best understanding	1(7%)	12(86%)	1(7%)
Q20: Most trustworthy	1(7%)	12(86%)	1(7%)

Some participants described relying primarily on the initial overview and only accessing additional explanations when something appeared unclear or unexpected. Others actively explored detailed explanations to understand how requirements were evaluated and how different course choices affected degree fulfillment.

Several participants expressed in Q19 and Q21 that the ability to access explanations incrementally was preferable to being presented with all available information at once. Progressive disclosure was described as supporting both efficiency and understanding, allowing participants to tailor the level of detail to their immediate needs. One participant noted that *"it was good to start with an overview and then choose to see more when something didn't make sense"* (P12).

At the same time, responses suggest that the appropriate level of explanation may vary depending on users' familiarity with degree requirements and their current planning situation. While some participants indicated in Q19 that they would prefer more detailed explanations by default, others expressed a preference for a more minimal presentation unless problems were detected.

These variations underscore the importance of flexibility in explanation design and provide empirical grounding for the system's approach to explanation granularity, in which explanation depth is made available through interaction rather than fixed in advance.

4.6 Summary of findings

In summary, the results indicate that participants generally perceived the system as supportive of understanding academic progression and degree requirements. Explanations played a central role in shaping participants' experiences, particularly when they made formal rules explicit and traceable in relation to individual course data.

Participants valued the combination of overview-level feedback and access to more detailed explanations, and many described the system as useful for preparing for academic planning decisions. At the same time, participants remained aware of the system's role as a decision support tool rather than a replacement for personal judgment or academic advising.

These findings provide empirical grounding for the system's design choices and motivate

the interpretive discussion of implications, limitations, and future directions presented in the following chapter.

5 Discussion

This chapter interprets the results of the evaluation in relation to the research questions and the design goals of the thesis. The discussion focuses on what the results mean for the design of rule-based decision support systems for academic study progression, and how explainability, correctness, and user trust interact in practice. The chapter concludes by reflecting on limitations and outlining directions for future work.

5.1 Interpretation of results in relation to the research questions

The main research question of this thesis asks how a rule-based decision support system for course progression can be designed so that students understand and trust its feedback regarding fulfillment of degree requirements. The evaluation results suggest that this can be achieved by combining formal correctness with explanation mechanisms that expose the relationship between student data and institutional rules, while preserving user agency.

A central finding is that users' understanding was strongly influenced by whether the system made degree requirements explicitly interpretable. Participants reported greater clarity when unmet requirements were not merely stated, but explained in terms of missing credits, prerequisites, or requirement categories. This supports the thesis assumption that correctness alone is insufficient; students must be able to see how conclusions are derived in order to form an accurate mental model of their progression.

Minimal versus detailed explanations

The results indicate that minimal feedback and more detailed explanations serve different but complementary purposes. Minimal feedback was generally perceived as useful for gaining a quick overview of progression status and identifying whether issues exist. However, when participants encountered unmet requirements or unexpected constraints, minimal feedback alone was often insufficient to support understanding and caused confusion.

Access to more detailed explanations allowed participants to inspect the causes behind system conclusions and reduced uncertainty about how requirements were evaluated. Importantly, the evaluation suggests that detailed explanations are most valuable when they are optional and revealed through interaction, rather than imposed by default. This supports the progressive disclosure approach adopted in the system design.

Rather than indicating that one explanation level is inherently superior, the results point to a trade-off: minimal explanations support efficiency and orientation, while detailed explanations support trust, transparency and learning. The system's ability to support both within a single interface was therefore central to its effectiveness.

Expert and student perspectives

The pilot study with the study counsellor and the student user study provided complementary perspectives. From the counsellor's viewpoint, the system's reasoning and explanations were largely aligned with professional advising practices, particularly in how degree requirements were interpreted and communicated. The counsellor's willingness to recommend the system as a preparatory tool suggests that the system's explanations were not perceived as misleading or overly authoritative.

Students, on the other hand, emphasized the value of transparency and the ability to independently explore their academic situation. While they trusted the system's assessment of formal requirements, they remained aware that personal preferences and life circumstances fall outside its scope. This alignment between expert caution and student expectations suggests that the system's positioning as a decision support tool—rather than a decision-maker—was appropriate and credible.

5.2 Answering the research question through interpretation of results

The main research question of this thesis asks how a rule-based decision support system for course progression can be designed so that students understand and trust its feedback regarding their fulfillment of degree requirements. The evaluation results suggest that this can be achieved by designing the system around three interdependent principles: formal correctness grounded in explicit rules, explainability through traceable reasoning, and interaction designs that preserve user agency.

First, the results indicate that **formal correctness alone is not sufficient**, but it is a necessary foundation for trust. Both students and the study counsellor treated the system's feedback as credible when it was clearly grounded in official degree requirements and prerequisite structures. The use of a rule-based reasoning approach enabled the system to produce conclusions that aligned with professional advising practice, which in turn supported users' willingness to rely on the system's assessments. This confirms that explicit rule representation is an appropriate basis for decision support in domains governed by formal regulations.

Second, the results show that **understanding and trust emerge when correctness is paired with explanation**. Participants reported greater clarity when the system exposed how course data was mapped to degree requirements, particularly in cases involving unmet area requirements or prerequisite constraints. Explanations allowed users to form an accurate mental model of the system's reasoning, transforming system output from an unclear judgment into an interpretable assessment. This finding directly addresses the research question by demonstrating that explainability is not an optional enhancement, but a central design requirement for trustworthy decision support.

Third, the evaluation demonstrates that **explanation depth must be flexible rather than fixed**. Minimal feedback was effective for providing orientation and quickly identifying potential issues, while more detailed explanations were necessary for diagnosing problems and resolving uncertainty. Importantly, trust did not arise from maximal transparency alone, but from the ability to access deeper explanations when needed. The progressive disclosure design supported this balance, allowing users to control explanation depth without overwhelming them. This suggests that systems designed to support understanding and trust should offer explanations as an interactive resource rather than a static presentation.

Finally, the results highlight the importance of **preserving user agency**. Participants consistently viewed the system as a support for reasoning rather than as an authority that should make decisions on their behalf. The presentation of multiple alternative study trajectories reinforced this perception by demonstrating that academic planning is not a single-path problem. This aligns with both student expectations and expert advising practice, and indicates that trust is strengthened when systems respect the user's role in decision-making rather than attempting to replace it.

Taken together, these findings provide a concrete answer to the research question: a rule-based decision support system can support student understanding and trust by ensuring formal correctness through explicit rules, exposing the reasoning behind system conclusions through structured explanations, and enabling users to explore this reasoning at varying levels of detail while retaining control over final decisions.

These results can also be interpreted in light of broader principles associated with Trustworthy AI, particularly those related to transparency, explainability, and human agency [4]. While the system developed in this thesis does not aim to fulfill or certify compliance with ethical AI frameworks, participants' responses reflect several of the concerns these principles address. Trust was found to be calibrated rather than unconditional, grounded in an understanding of both the system's reasoning and its limitations. This suggests that principles associated with Trustworthy AI can serve as a useful conceptual lens for interpreting user experiences of rule-based decision support systems, even in non-automated contexts such as academic study planning.

5.3 Reflections on technical choices

An important design decision in this thesis was the choice of a rule-based reasoning approach rather than a data-driven or machine learning–based solution. This choice was motivated by several factors.

First, degree requirements are formal, explicitly defined, and relatively stable. Modeling them using declarative rules aligns naturally with their structure and avoids the need for large datasets or training processes that are unnecessary for this domain. Second, rule-based systems support transparency and traceability, which are essential for explainability and trust. Unlike many machine learning approaches, the reasoning process can be inspected and explained in terms that align with institutional regulations.

Privacy considerations also support this choice. A rule-based system can operate on well-defined academic records without requiring aggregation of large amounts of personal data or behavioral tracking, reducing privacy risks and ethical concerns.

The absence of formally structured degree requirements in existing systems, such as Ladok, posed a practical challenge. Degree requirements are typically published as textual descriptions or documents on university websites rather than as coded or automated regulations in the system. This limitation in this thesis was addressed by focusing on a single programme as a proof of concept and manually extracting and formalizing its requirements into structured representations. While this approach does not scale directly, it demonstrates that formalization is feasible and highlights a broader infrastructural gap in current academic systems.

5.4 Limitations

This thesis has several limitations that should be acknowledged.

First, the user study involved a limited number of participants from a single programme. While the results provide valuable insight into user experience and perception, they are not statistically generalizable. The findings should therefore be interpreted as exploratory rather than definitive.

Second, the system's reasoning model covers a subset of academic rules relevant to the selected programme. Although sufficient for demonstrating the approach, the model does not capture the full diversity of degree structures, exceptions, or institutional variations found across programs and universities.

5.5 Future work

One important direction is the personalization of planning objectives. While this thesis deliberately avoids optimizing for individual preferences, future work could explore how formal correctness can be combined with soft constraints representing workload balance, pacing preferences, or career goals—without undermining transparency.

A further area for development is institutional integration. If degree requirements were made available in structured, machine-readable formats, rule-based reasoning engines could be scaled to support multiple programs and universities. Integration with real student information systems would also enable longitudinal evaluation of how decision support affects planning behavior over time.

Finally, future studies could investigate long-term use, including how students' understanding and trust evolve with repeated interaction, and how such systems might support collaboration between students and study counsellors rather than replacing human guidance.

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A Appendix

A.1 Complete Answer Set Programming Rule Set (reasoning engine)

This appendix contains the complete Answer Set Programming (ASP) rule set used by the reasoning engine described in Chapter 3. The rules encode course prerequisites, credit accumulation, degree requirements, and optimization criteria. They are included to support transparency and reproducibility of the system's reasoning, but are not required reading for understanding the system design.

```
pool(C,T) :- passed(S,C), offered(C,T).
pool(C,T) :- take(C,T).
pool_course(C) :- pool(C,_).

1 { take(C,T) : offered(C,T) } 1 :-
    course(C,_,_,_),
    student(S),
    not passed(S,C),
    not unfulfillable(C).

term_before(T1,T2) :-
    term_order(T1,O1),
    term_order(T2,O2),
    O1 < O2.

prereq_ok(C,P) :-
    prereq_course(C,P),
    passed(_,P).

prereq_ok(C,P) :-
    prereq_course(C,P),
    take(C,TC),
    take(P,TP),
    term_before(TP,TC).

prereq_fail(C,P) :-
    prereq_course(C,P),
    not prereq_ok(C,P).

unfulfillable(C) :-
    prereq_fail(C,_).

:- take(C,_), unfulfillable(C).
```

```
total_units_before(Term,Sum) :-
    term(Term),
    Sum = #sum {
        Hp,C,T2 :
            pool(C,T2),
            course(C,Hp,_,_),
            term_before(T2,Term)
    }.

% can the requirement ever be met?
can_meet_credit_req(C,Req) :-
    prereq_min_credits(C,Req),
    offered(C,TermC),
    total_units_before(TermC,Have),
    Have >= Req.

% fail if no possible term meets the requirement
prereq_credit_fail(C,Req) :-
    prereq_min_credits(C,Req),
    not can_meet_credit_req(C,Req).

unfulfillable(C) :-
    prereq_credit_fail(C,_).

area_units_before(A,Term,Sum) :-
    course_area(_,A),
    term(Term),
    Sum = #sum {
        Hp,C,T2 :
            pool(C,T2),
            course(C,Hp,_,_),
            course_area(C,A),
            term_before(T2,Term)
    }.

can_meet_area_req(C,A,Req) :-
    prereq_min_area_credits(C,A,Req),
    offered(C,TermC),
    area_units_before(A,TermC,Have),
    Have >= Req.

prereq_area_fail(C,A,Req) :-
    prereq_min_area_credits(C,A,Req),
    not can_meet_area_req(C,A,Req).

unfulfillable(C) :-
    prereq_area_fail(C,_,_).
```

```
term_units(Term,Sum) :-
    term(Term),
    Sum = #sum {
        Hp,C :
            pool(C,T2),
            take(C,Term),
            course(C,Hp,_,_)
    }.

:- term_units(Term,Sum), Sum > 320.

deviation(Term,Dev) :-
    term_units(Term,Sum),
    Dev = |Sum - 300|.

#minimize { Dev@2,Term : deviation(Term,Dev) }.

active_term(T) :- take(_,T).

slack(T,Slack) :-
    active_term(T),
    term_units(T,Sum),
    Slack = 320 - Sum.

#minimize { 1@4, T : active_term(T) }.
#minimize { Slack@3, T : slack(T,Slack) }.
#maximize { Hp@2, C, T : take(C,T), course(C,Hp,_,_) }.

total_units(Sum) :-
    Sum = #sum { Hp,C : pool_course(C), course(C,Hp,_,_) }.

area_units(A,Sum) :-
    goal_area(A,_),
    Sum = #sum {
        Hp,C :
            pool_course(C),
            course(C,Hp,_,_),
            course_area(C,A)
    }.

goal_total_fail(Req,Have) :-
    goal_total(Req),
    total_units(Have),
    Have < Req.

goal_area_fail(A,Req,Have) :-
    goal_area(A,Req),
```

```
    area_units(A,Have),
    Have < Req.

exam_fail_total :- goal_total_fail(_, _).
exam_fail_area(A) :- goal_area_fail(A,_,_).
#minimize { 1@4,C : take(C,_) }.

#minimize { 1@1 : exam_fail_total }.
#minimize { 1@1,A : exam_fail_area(A) }.

#show take/2.
#show unfulfillable/1.
#show goal_total_fail/2.
#show goal_area_fail/3.
#show prereq_fail/2.
#show prereq_area_fail/3.
```

A.2 User Study Questionnaire

This questionnaire was used in the user study described in Chapter 4. The questions were administered after participants had explored the prototype for the decision support system. Responses were anonymous. Likert-scale questions used a five-point scale unless otherwise stated. The questionnaire was administered in Swedish, as all participants were Swedish speakers. To preserve the exact wording presented to participants, the original Swedish version of the questionnaire is included below. In the main text, questionnaire items are referred to by question number and described in English.

A.2.1 Samtycke

Q1. Ger du ditt samtycke till att delta i studien? ( ) Ja ( ) Nej

A.2.2 Bakgrundsinformation

Q2. Hur gammal är du?

- 19-24
- 25-29
- 30-35
- Annat

Q3. Vilket kön identifierar du dig med?

- Man
- Kvinna

- Icke-binär
- Föredrar att inte säga

Q4. Vilket program läser du?

Q5. Har du tidigare använt något digitalt system för att följa eller planera din studieprogression?

- yes
- No

Q6. Hur trygg känner du dig generellt i din förståelse av examenskraven för ditt program? (1) Mycket otrygg - (5) Mycket trygg

A.2.3 Upplevd förståelse

Q7. Med egna ord: hur uppfattade du systemets bedömning av studieprogressionen?

Q8. Upplevde du att systemet tydligt förklarade varför examenskraven var uppfyllda eller inte?
(1) Inte alls - (5) Mycket tydligt

Q9. Vilka delar av systemets återkoppling hjälpte dig mest att förstå situationen i förhållande till examenskraven?

A.2.4 Förklaringar och informationsnivå

Q10. Hur upplevde du balansen mellan översiktlig och detaljerad information i systemet? (1) För lite information - (5) För mycket information

Q11. Fanns det någon information som kändes onödig, svår att tolka eller förvirrande?

Q12. I vilka situationer upplevde du att de mer detaljerade förklaringarna var särskilt hjälpsamma?

A.2.5 Tillit till systemet

Q13. I vilken utsträckning litade du på systemets bedömning av studieprogressionen? (1) Inte alls - (5) I mycket hög utsträckning

Q14. Fanns det något tillfälle då du tvivlade på systemets slutsatser? I så fall, varför?

Q15. Vad, om något, gjorde att du kände ökad tillit till systemet?

A.2.6 Upplevd kontroll och handlingsförmåga

Q16. Upplevde du att systemet gav dig tillräcklig information för att förstå vad du kan eller bör göra härnäst i dina studier?

(1) Inte alls - (5) I mycket hög grad

Q17. Kände du att systemets återkoppling gav dig en känsla av kontroll över din studieprogression?

A.2.7 Jämförelse mellan översiktlig och detaljerad återkoppling

Q18. Vilken version upplevde du gav bäst förståelse för din studieprogression och examenskraven?

- Den översiktliga (minimala) versionen
- Den detaljerade versionen
- Ingen tydlig skillnad

Q19. Varför upplevde du denna version som mer begriplig?

Q20. Vilken version kände du störst tillit till?

- Den översiktliga (minimala) versionen
- Den detaljerade versionen
- Ingen tydlig skillnad

Q21. Varför upplevde du större tillit till denna version?

A.2.8 Helhetsintryck och reflektion

Q22. Om du skulle använda ett sådant system i verkligheten, vad skulle du vilja ändra eller förbättra?

Q23. Hur skulle du beskriva systemets återkoppling med ett eller två ord?

A.3 User interface documentation

This appendix provides screenshots of the implemented user interface to document how the system is presented to users during evaluation. The screenshots illustrate the main interaction views relevant to the study, including the start page, the study progression overview, and the minimal and detailed explanation modes.

The purpose of this appendix is not to analyse the interface design in detail, but to support transparency and reproducibility by showing how the system's functionality and explanations were concretely presented to participants during the user study. All screenshots correspond to the version of the system used in the evaluation described in Chapter 4.

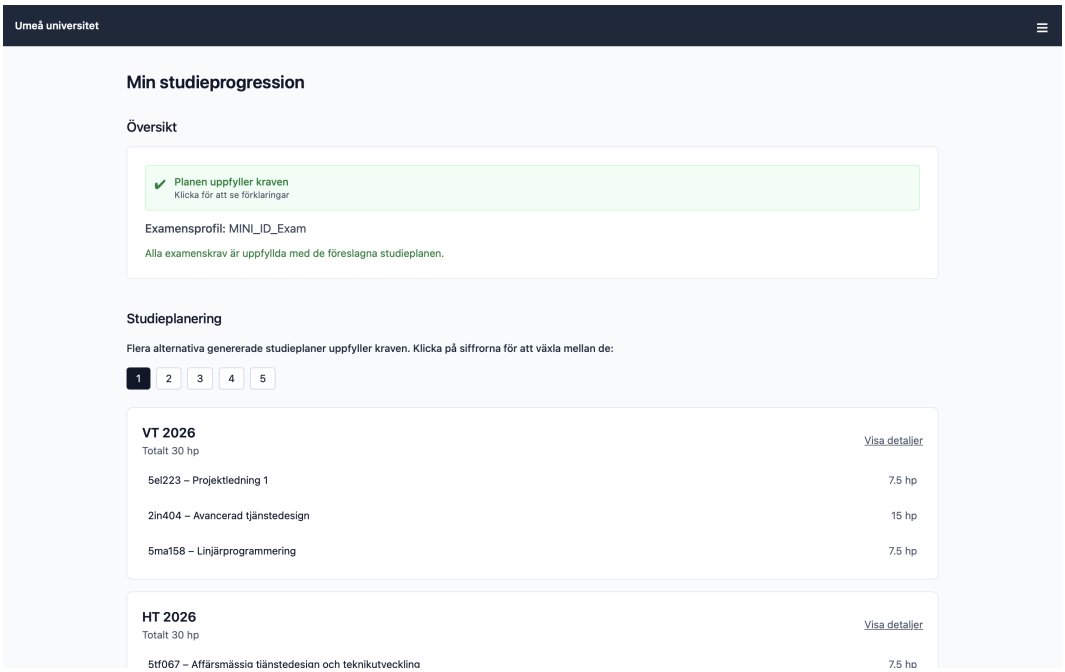


Figure 2: Initial view of the study progression interface when all degree requirements are fulfilled, presenting minimal, high-level feedback and multiple alternative study trajectories for user exploration.

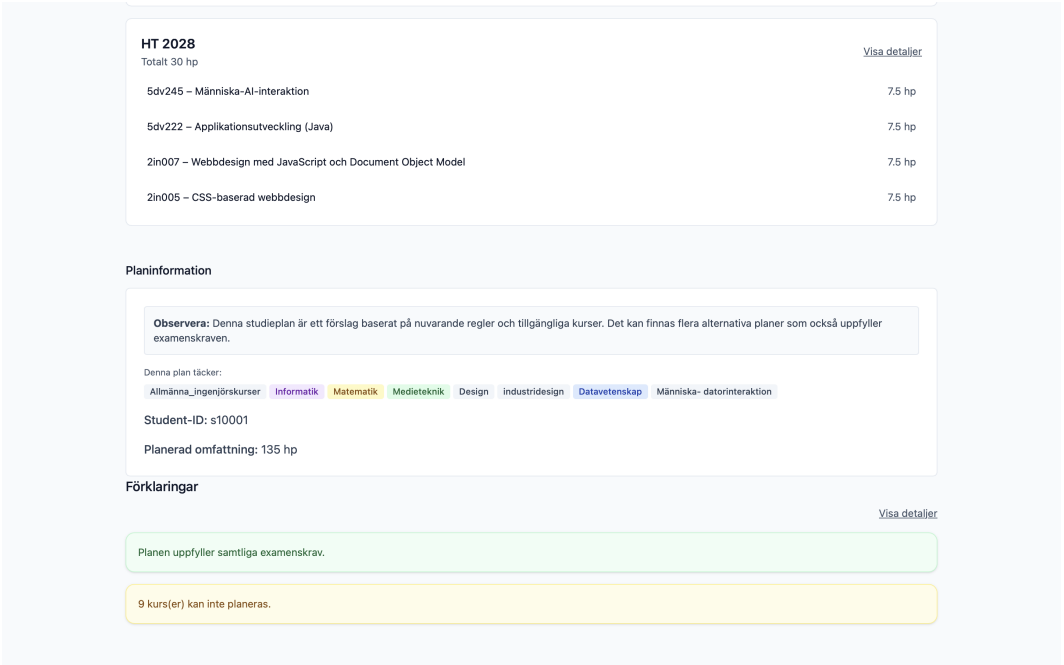


Figure 3: Lower section of the study progression interface when all degree requirements are fulfilled, revealed through scrolling, continuing the minimal feedback presentation.

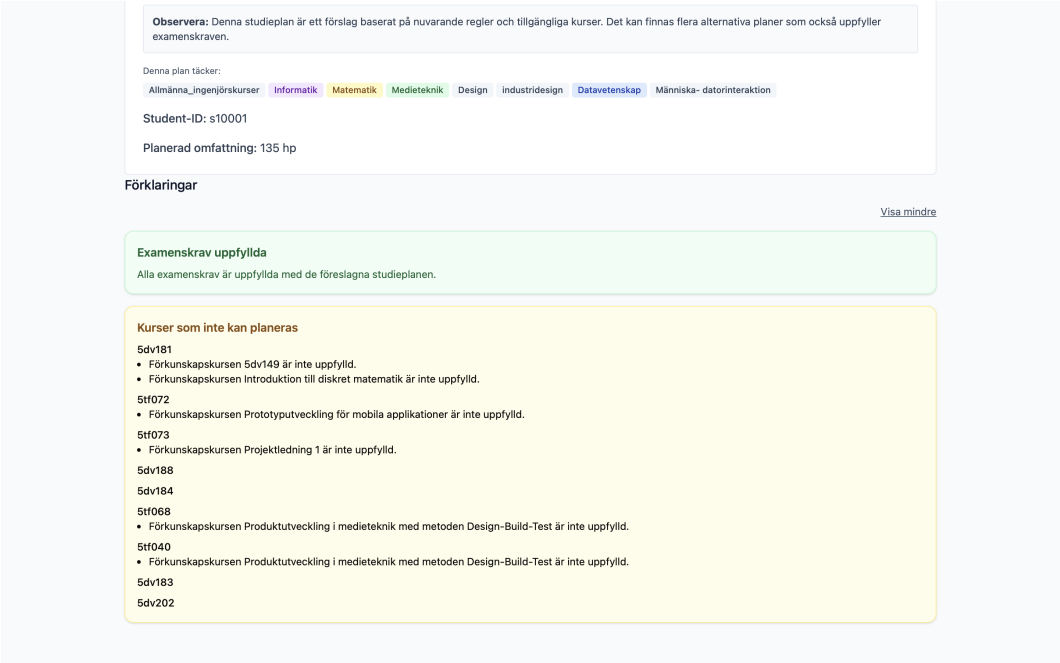


Figure 4: Study progression interface in detailed explanation mode when all degree requirements are fulfilled, showing expanded feedback and courses that could not be included in the proposed plans.

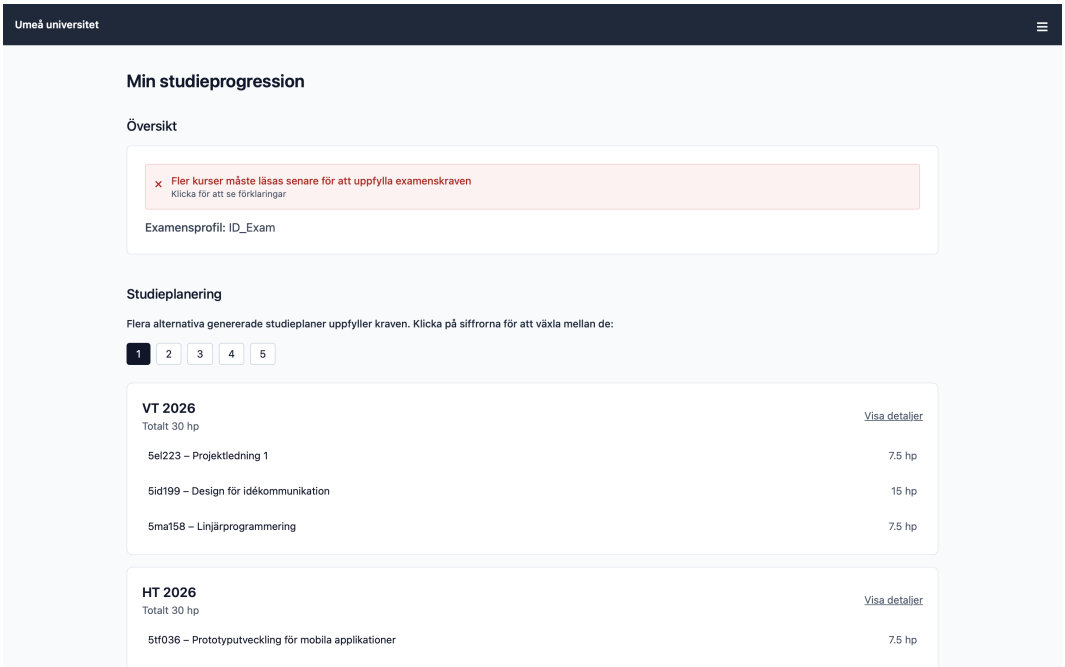


Figure 5: Initial view of the study progression interface when degree requirements are not fulfilled, presenting minimal feedback and multiple alternative study trajectories.

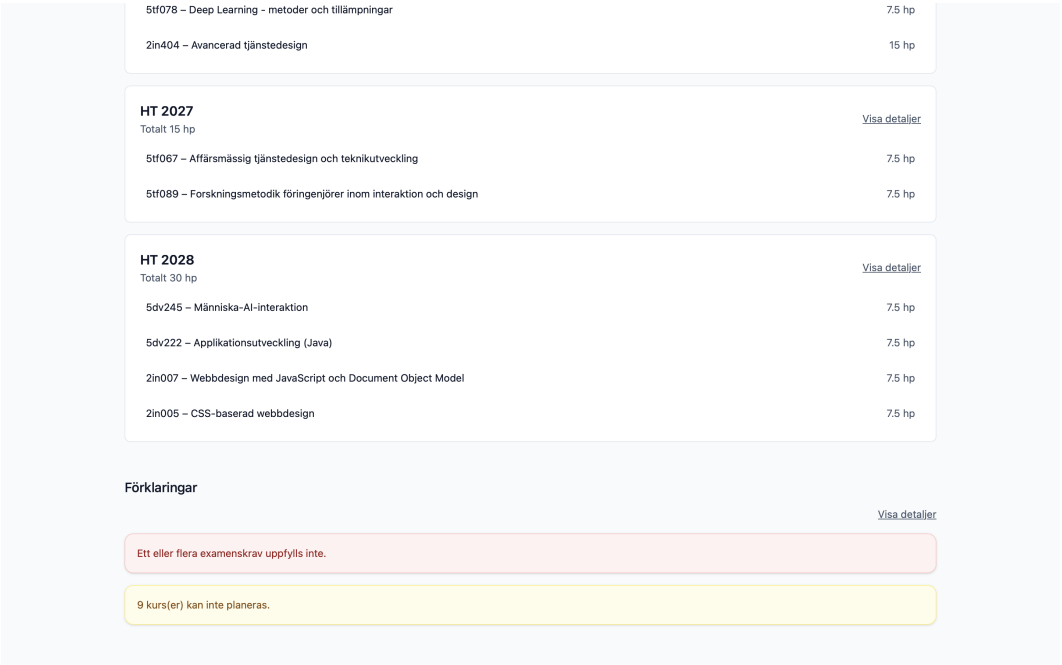


Figure 6: Lower section of the study progression interface when degree requirements are not fulfilled, revealed through scrolling, maintaining a minimal feedback presentation.

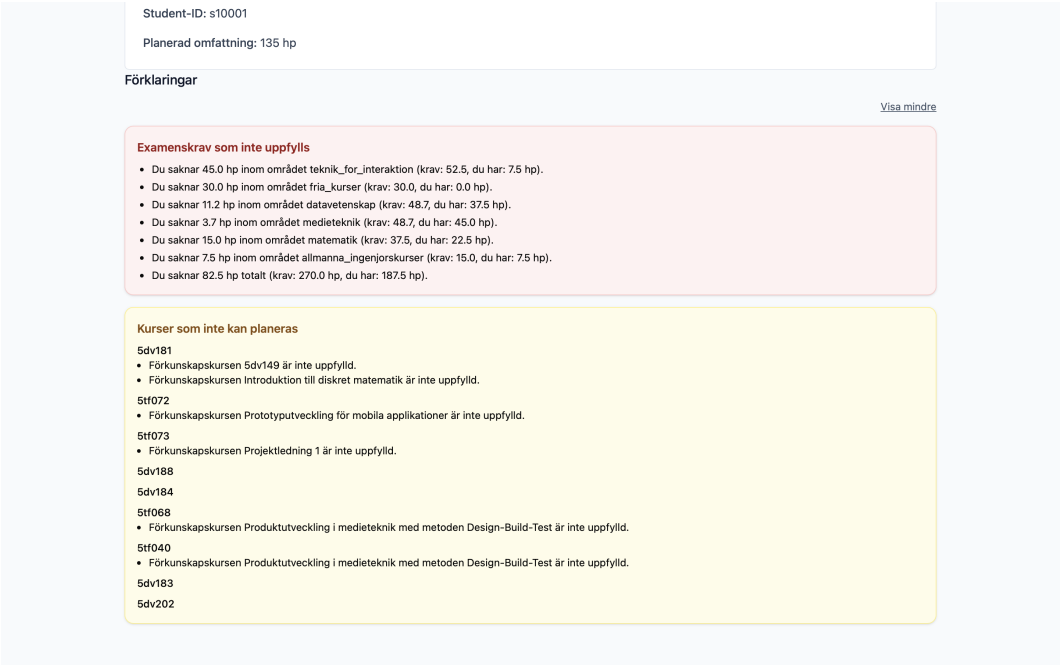


Figure 7: Study progression interface in detailed explanation mode when degree requirements are not fulfilled, showing explanatory feedback for the requirements and courses that could not be planned.

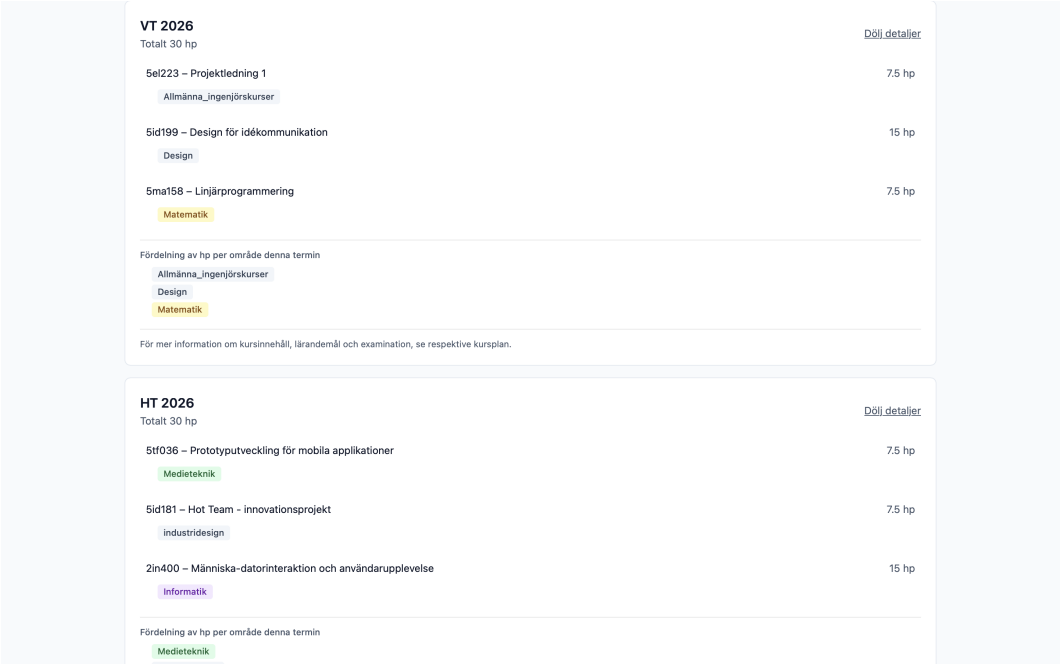


Figure 8: Detailed course planning view, illustrating how planned courses are distributed across academic terms and presented for user inspection.